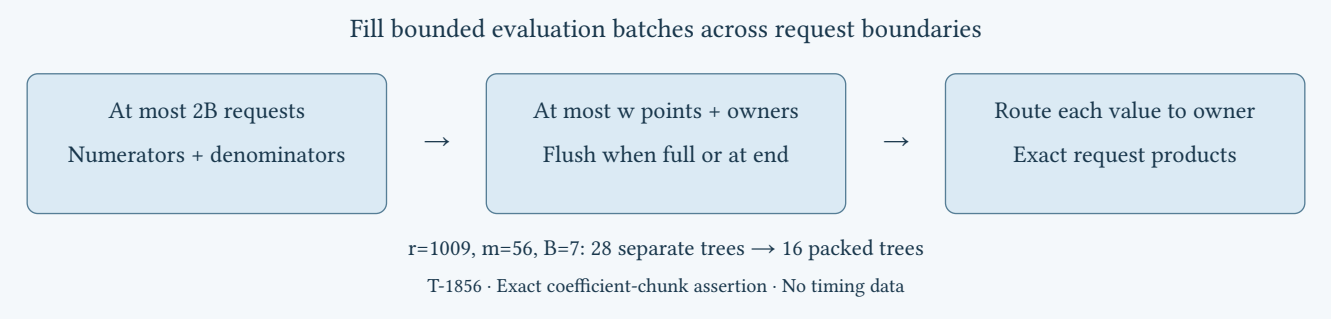
Share evaluation batches across recurrence products

T-1856 · 2026-11-07 · Bounded request packing in prime-field modes

The coefficient stream now submits numerator and denominator intervals together. Their block points fill one bounded buffer across request boundaries, carrying destination indices. A flush evaluates the cached polynomial and multiplies each value into its designated request. Short products and tails retain exact semantics.



Routing invariant. For each request, its accumulator contains precisely its direct tail and already-flushed block factors; pending point/owner pairs identify the rest. Each pair is consumed once. Repeated points, zero factors and wrap are admitted because monic remainder-tree evaluation does not need distinct-point inverses. All requests are validated before helper work.

Exact tree count. If blocked request i contributes t_i points, separate scheduling needs sum_i ceil(t_i/w) trees. Packing needs ceil(sum_i t_i/w), never more. It changes neither the number of block values nor the product factors. For seven transitions at r=1009, m=56, B=7, fourteen requests each contribute eight points:28separate trees become16packed trees. Exact coefficient equality is asserted.

Deterministic native check	Count	Outcome
Request-list panels	375	3,195 exact request products
Fewer / equal tree panels	123 / 252	Both count formulas hold
Real coefficient chunk	16 vs28	Same emitted coefficients
Inherited interval checks	4,820	Zero/wrap controls pass
Public m56 panel	281	14,421 calls; zero unresolved
Inherited public panel	617	5,929 calls; zero unresolved

Costs and limits. At most2Brequests/results, wlive points and owners, one cached degree-w polynomial and O(wlogw) tree coefficients; allocator/kernel workspace is separate. No complete block-point grid is stored. Fewer trees improves the conservative per-batch upper bound, but exact kernel arithmetic depends on tree sizes. No field-operation total or timing dominance is asserted. Existing source, first-C0, order-validation and inversion costs remain.

Evidence. validation01 pins47transitive inputs and stdout; native overflow checks enabled. Coefficient/residue/state controls also pass. PCC-20261107-720c23 is unsigned producer evidence. No native/report failure, cloud trial or fullCargo rerun. Original4,563-case cohort and prospective trace protocol unchanged.

Sources and next step. Exact predecessor and Bostan-Gaudry-Schost2004 standard multipoint evaluation; KB unavailable. This is request scheduling, not specialized progression interpolation. Next investigate structured progression evaluation or repeated block-value reuse with explicit caps and preconditions. DERIVATION.md gives proof, exact replay and limits. Historical/canonical graphs remain unchanged.